

Energy

5.1 INTRODUCTION

Energy is the capacity to do work, produce motion or force or carry out transformations. It can be in different forms like thermal energy, electrical energy, mechanical energy or chemical energy. The raw form in which the energy resources occur in nature are the **primary energy resources** (coal, petroleum, natural gas, wind, solar) which are converted into some intermediate form (steam or chemicals) that are finally converted into usable or **secondary energy** (fuels, electricity etc.). Every development activity depends on energy. Therefore, energy resources and their usage directly influence national economy and growth of civilisation. While energy requirements of primitive man were limited, the energy needs in modern times have tremendously increased due to fast development, increasing transportation, industrialisation, raised standards of living and rapid population growth.

Energy consumption of a nation is usually considered as an index of its development. This is because almost all the developmental activities are directly or indirectly dependent upon energy. We find wide disparities in per capita energy use between the developed and the developing nations.

The first form of energy technology probably was the fire, which produced heat and the early man used it for cooking and heating purposes. Wind and hydropower have also been in use for the last 10,000 years. The invention of steam engines replaced the burning of wood by coal and coal was later replaced to a great extent by oil. In 1970's due to Iranian revolution and Arab oil embargo the prices of oil shot up. This ultimately led to exploration and use of several alternate sources of energy.

5.2 TYPES OF ENERGY RESOURCES

All energy resources are available in nature in their raw form, which are explored, extracted, processed and converted into final usable forms.

Based upon their usage over a long time, energy resources are generally classified as conventional and non-conventional. Demarcation between the two is not so rigid, because what is a non-conventional form today, may in due course of time, become conventional.

- **Conventional Energy Resources:** Energy resources that are traditionally in use for all these years (particularly prior to the oil crisis of 1973) are known as conventional forms, *e.g.* coal, petroleum, natural gas, fire-wood, hydropower and even nuclear fission fuels.
- **Non-conventional Energy Resources:** These include the alternate resources of energy that are being considered and commercialised for large scale use after the oil crisis. These resources are going to have increased share of energy use in future. These alternate resources are generally renewable forms including solar, wind, geothermal, ocean wave, tidal, biomass, biogas, nuclear fusion energy etc.

A more appropriate categorisation of an energy resource is based on its durability and regenerating capacity, which classifies it as renewable or non-renewable.

- **Renewable Resources are those which can be generated continuously in nature and are inexhaustible** *e.g.* wood, solar energy, wind energy, tidal energy, hydropower, biomass energy, bio-fuels, geo-thermal energy and hydrogen. They are also known as non-conventional sources of energy and they can be used again and again in an endless manner.
- **Non-renewable Resources which have accumulated in nature over a long span of time and cannot be quickly replenished when exhausted** *e.g.* coal, petroleum, natural gas and nuclear fuels like uranium and thorium.

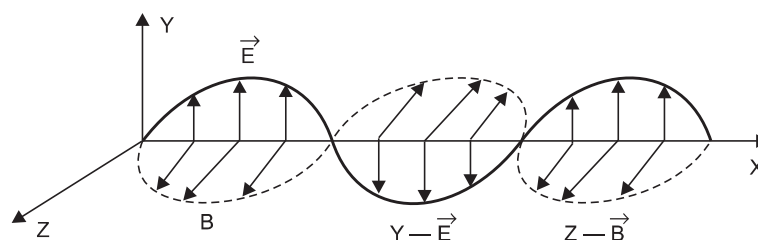
Wood is a renewable resource as we can get new wood by growing a sapling into a tree within 15–20 years but it has taken millions of years for the formation of coal from trees and cannot be regenerated in our life time, hence coal is not renewable.

5.3 ELECTROMAGNETIC RADIATIONS

Life on earth has evolved in an environment that has several types of electromagnetic radiations coming from sources like thunderstorms, lightening, solar radiations, etc. Human beings have also added electromagnetic radiations in several ways through high frequency electromagnetic radiations (EMR) such as microwaves, X-rays, electric power systems and distribution systems, transportation, electric gadgets, mobile phones, sound systems and so on. The electromagnetic radiations have increased more than a million times in the environment due to human activities as compared to natural electromagnetic radiations.

Properties of Electromagnetic Radiations

- Electromagnetic radiation (waves) can travel in space without any material medium.
- There is sinusoidal variation in electric field and similar in magnetic field, at right angles to each other.
- Both electric field ($\vec{E}$) and magnetic field ($\vec{B}$) vary with time and space and have the same frequency, as shown in Fig. 5.1.



X-direction of propagation of electromagnetic wave.

Fig. 5.1. Electromagnetic field.

- Both electric and magnetic field components coexist in time and space.
- Examples of electromagnetic radiation are radiowaves infrared rays, X-rays, γ -rays, microwaves.
- Electromagnetic radiations have a wide range of wavelength. Table 5.1 shows the wavelength and sources of various types of these waves.

Table 5.1. Range of Different Electromagnetic Radiations

Type	Wavelength (m)	Source
Radiowaves	0.3 to 6×10^2	Oscillating circuit
Microwaves	10^{-3} to 0.3	Oscillating current in special vacuum tube
Infra-red	8×10^{-7} to 1×10^{-3}	Excitation of atoms and molecules
Visible light	4×10^{-7} to 8×10^{-7}	Excitation of valency electrons.
Ultra violet	6×10^{-10} to 4×10^{-7}	Excitation of atom, spark and arc lamp
X-rays	1×10^{-13} to 3×10^{-8}	Sudden deceleration of high energy electrons.
Gamma rays	6×10^{-14} to 1×10^{-10}	Nuclear origin

Different electromagnetic radiations are of different use. While the visible range light is responsible for photosynthesis by green plants, which forms the primary productivity—the base of the food chain. Other types of radiations find use in various sectors of transportation, communication, medical and scientific research.

Exposure of living organisms including human beings to some of these electromagnetic radiations has become a cause of alarm, as these waves have been reported to have several harmful effects on health, including increased chances of tumor formation, adverse reproductive effects and behavioral and neurological disorders.

5.4 CONVENTIONAL ENERGY RESOURCES (FOSSIL FUEL BASED)

These are the fossil fuels like coal, petroleum, natural gas and nuclear fuels. These were formed by the decomposition of the remains of plants and animals buried under the earth millions of years ago. These fuels are very precious because they have taken a long time to be formed and if we exhaust their

reserves at such a fast rate as we have been doing, ever since we discovered them, then very soon we will lose these resources forever.

5.4.1 Coal

Coal was formed 255–350 million years ago in the hot, damp regions of the earth during the carboniferous age. The ancient plants along the banks of rivers and swamps were buried after death into the soil and due to the heat and pressure gradually got converted into peat and coal over millions of years. There are mainly three types of coal, namely *anthracite* (hard coal), *bituminous* (soft coal) and *lignite* (brown coal). Anthracite coal has maximum carbon (90%) and calorific value (8700 kcal/kg.). Bituminous, lignite and peat contain 80, 70 and 60% carbon, respectively. Coal is the most abundant fossil fuel in the world. *At the present rate of usage, the coal reserves are likely to last for about 200 years and if its use increases by 2% per year, then it will last for another 65 years.*

India has about 5% of world's coal and Indian coal is not very good in terms of heat capacity. Major coal fields in India are Raniganj, Jharia, Bokaro, Singrauli and Godavari valley. The coal states of India are Jharkhand, Orissa, West Bengal, Madhya Pradesh, Andhra Pradesh and Maharashtra. Anthracite coal occurs only in J & K.

When coal is burnt it produces carbon dioxide, which is a greenhouse gas responsible for causing enhanced global warming. Coal also contains impurities like sulphur and as it burns the smoke contains toxic gases like oxides of sulphur and nitrogen. Coal conversion technologies involve conversion of coal from solid form to liquid or gaseous form by *coal liquefaction* and *gasification*, respectively. Direct burning of coal releases emissions like smoke, particulate matter, SO_x , NO_x , CO and CO_2 , whereas gaseous or liquid fuel forms cause less pollution. Energy from petroleum is harnessed by refining and fractional distillation.

5.4.2 Petroleum

It is the lifeline of global economy. There are 13 countries in the world having 67% of the petroleum reserves which together form the OPEC (Organization of Petroleum Exporting Countries). About 1/4th of the oil reserves are in Saudi Arabia.

At the present rate of usage, the world's crude oil reserves are estimated to get exhausted in just 40 years. Some optimists, however, believe that there are some yet undiscovered reserves. Even then the crude oil reserves will last for another 40 years or so. Crude petroleum is a complex mixture of alkane hydrocarbons. Hence, it has to be purified and refined by the process of **fractional distillation**, during which process different constituents separate out at different temperatures. We get a large variety of products from this, namely, petroleum gas, kerosene, petrol, diesel, fuel oil, lubricating oil, paraffin wax, asphalt, plastic etc.

Basic steps in harnessing energy from petroleum include exploration of oil resources, drilling of wells, production, storage and transport of crude oil, refining of crude oil, storage and transportation of products.

Exploration is usually done in areas having sedimentary rocks, where petroleum deposits generally occur, which may be on-shore (land) or off-shore (sea) at shallow, medium or large depths.

Drilling is done from specifically formed drilling platforms. For enhanced extraction, the following techniques are used:

- By injecting fluid (air, gas, steam, water) into the well.
- By using chemical explosives to loosen tight formations.
- By adding chemicals to reduce viscosity of the crude oil.
- By allowing microbial growth inside to increase bulk, reduce viscosity and enhance recovery, known as Microbially Enhanced Oil Recovery (MEOR).
- By controlled underground burning to push up oil.

After extraction, crude oil is separated from natural gas and water, stored and transported through pipelines or tankers.

Crude oil is refined by the following processes:

- Separation of some components by distillation.
- Chemical purification or removal of impurities by adsorption (on charcoal).
- Formation of hydrocarbons by cracking or hydrogenation.

During refining of crude oil, several products are obtained that are used in domestic, transport, industrial and electric power sectors.

Petroleum is a cleaner fuel as compared to coal as it burns completely and leaves no residue. It is also easier to transport and use. That is the reason why petroleum is preferred amongst all the fossil fuels.

Liquefied Petroleum Gas (LPG): The main component of petroleum is butane, the other being propane and ethane. The petroleum gas is easily converted to liquid form under pressure as LPG. It is odourless, but the LPG in our domestic gas cylinders gives a foul smell. This is, in fact, due to ethyl mercaptan, a foul smelling gas, added to LPG so that any leakage of LPG from the cylinder can be detected instantaneously.

Oil fields in India are located at Digboi (Assam), Gujarat Plains and Bombay High, offshore areas in deltaic coasts of Godavari, Krishna, Kaveri and Mahanadi.

5.4.3 Natural Gas

It is mainly composed of methane (95%) with small amounts of propane and ethane. It is a fossil fuel. Natural gas deposits mostly accompany oil deposits because it has been formed by decomposing remains of dead animals and plants buried under the earth. **Natural gas is the cleanest fossil fuel.** It can be easily transported through pipelines. It has a high calorific value of about 50 kJ/g and burns without any smoke.

Currently, the amount of natural gas deposits in the world are of the order of $80,450 \text{ g m}^{-3}$. Russia has maximum reserves (40%), followed by Iran (14%) and USA (7%). Natural gas reserves are found in association with all the oil fields in India. Some new gas fields have been found in Tripura, Jaisalmer, off-shore area of Mumbai and the Krishna-Godavari Delta.

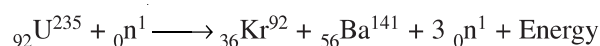
Natural gas is used as a domestic and industrial fuel. It is used as a fuel in thermal power plants for generating electricity. It is used as a source of hydrogen gas in fertilizer industry and as a source of carbon in tyre industry.

- **Compressed Natural Gas (CNG):** It is being used as an alternative to petrol and diesel for transport of vehicles. Delhi has totally switched over to CNG where buses and auto-rickshaws run on this new fuel. CNG use has greatly reduced vehicular pollution in the city.
- **Synthetic Natural Gas (SNG):** It is a mixture of carbon monoxide and hydrogen. It is a connecting link between a fossil fuel and substituted natural gas. Low grade coal is initially transformed into synthetic gas by gasification followed by catalytic conversion to methane.

5.4.4 Nuclear Energy

Nuclear energy is known for its high destructive power as evidenced from nuclear weapons. The nuclear energy can also be harnessed for providing commercial energy. Nuclear energy can be generated by two types of reactions:

(i) *Nuclear fission:* It is the nuclear change in which nucleus of certain isotopes with large mass numbers are split into lighter nuclei on bombardment by neutrons and a large amount of energy is released through a chain reaction as shown in Fig. 5.2



Nuclear reactors make use of nuclear chain reaction. In order to control the rate of fission, only 1 neutron released is allowed to strike for splitting another nucleus. Uranium-235 nuclei are most commonly used in nuclear reactors.

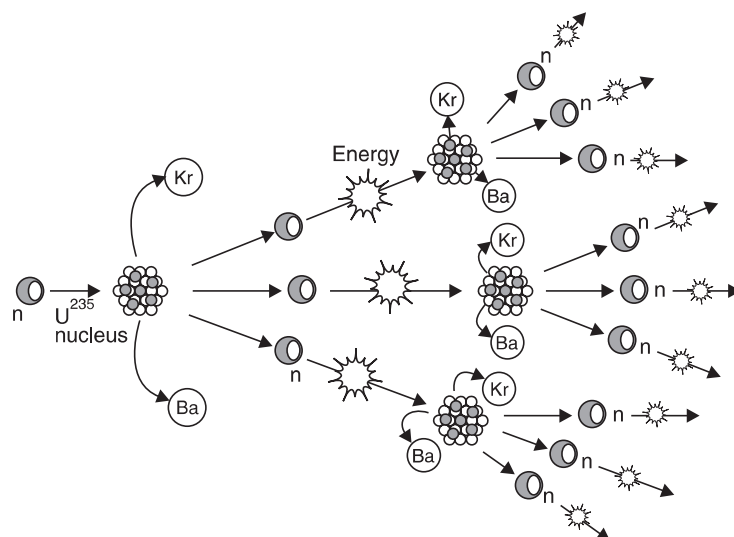


Fig. 5.2. Nuclear fission—a chain reaction initiated by one neutron that bombards a Uranium (U^{235}) nucleus, releasing a huge quantity of energy, two smaller nuclei (Ba, Kr) and 3 neutrons.

(ii) *Nuclear fusion*: Here two isotopes of a light element are forced together at extremely high temperatures (1 billion °C) until they fuse to form a heavier nucleus releasing enormous energy in the process. It is difficult to initiate the process but it releases more energy than nuclear fission (Fig. 5.3).

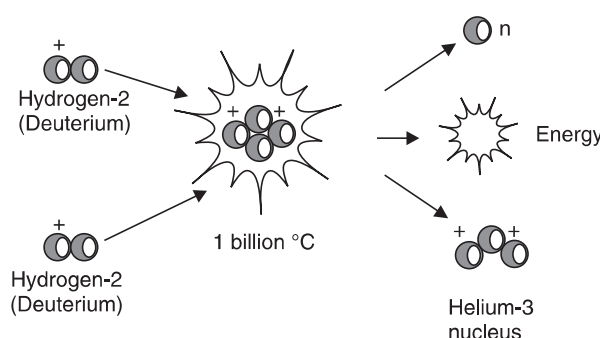


Fig. 5.3. Nuclear fusion reaction between two hydrogen-2 nuclei, which takes place at a very high temperature of 1 billion °C; one neutron and one fusion nucleus of helium-3 is formed along with a huge amount of energy.



Two hydrogen-2 (Deuterium) atoms may fuse to form the nucleus of Helium at 1 billion °C and release a huge amount of energy. Nuclear fusion reaction can also take place between one Hydrogen-2 (Deuterium) and one Hydrogen-3 (Tritium) nucleus at 100 million °C forming Helium-4 nucleus, one neutron and a huge amount of energy.

Nuclear energy has tremendous potential but any leakage from the reactor may cause devastating nuclear pollution e.g. the Chernobyl nuclear disaster (1986). Disposal of the nuclear waste is also a big problem.

Nuclear power in India is still not very well developed. The power plants in India are Pressurised Heavy Water Reactor (PHWR) Power plant type. The nuclear power plants are located at Tarapur (Maharashtra), Rana Pratap Sagar near Kota (Rajasthan), Kalpakkam (Tamil Nadu), Narora (U.P.), Kakrapar (Gujarat), Kaiga (Karnataka), Rawatbhata (Rajasthan) and Kudankulam (Tamil Nadu).

5.5 NON-CONVENTIONAL ENERGY SOURCES

5.5.1 Hydroelectric (Hydropower) Energy

The water flowing in a river is collected by constructing a big dam where the water is stored and allowed to fall from a height. Potential energy stored in water reservoir at high head is converted into kinetic energy in the flowing water. The blades of the turbine located at the bottom of the dam move with the fast moving water which in turn rotate the generator and produce electricity. We can also construct mini or micro hydropower plants on the rivers in hilly regions for harnessing the hydro energy on a small scale, but the minimum height of the waterfalls should be 10 metres. **The hydropower**

potential of India is estimated to be about 4×10^{11} kW hours. Till now we have utilised only a little more than 11% of this potential.

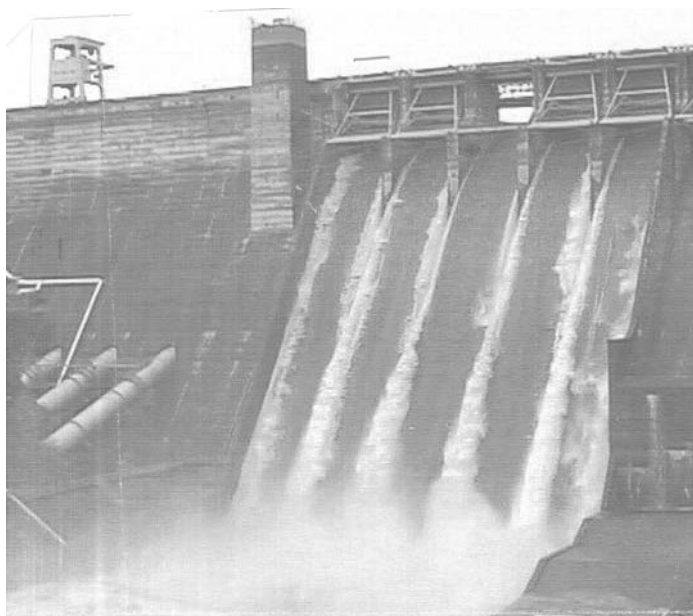


Fig. 5.4. Hydroelectric power (Dam).

Hydropower does not cause any pollution. It is renewable and normally the hydropower projects are multi-purpose projects that help in controlling floods, used for irrigation, navigation etc. However, big dams are often associated with a number of environmental impacts.

5.5.2 Solar Energy

Sun is the ultimate source of energy, directly or indirectly for all other forms of energy. The nuclear fusion reactions occurring inside the sun release enormous quantities of energy in the form of heat and light. The solar energy received by the near earth space is approximately $1.4 \text{ kilojoules/second/m}^2$ known as solar constant.

Traditionally, we have been using solar energy for drying clothes and food-grains, preservation of eatables and for obtaining salt from sea-water. Now we have several techniques for harnessing solar energy. Some important solar energy harvesting devices are discussed here.

(i) **Solar heat collectors:** These can be passive or active in nature. Passive solar heat collectors are natural materials like stones, bricks etc. or material like glass which absorb heat during the day time and release it slowly at night. Active solar collectors pump a heat absorbing medium (air or water) through a small collector which is normally placed on the top of the building.

(ii) **Photovoltaic cells (Solar cells) for direct conversion of solar radiations into electrical energy:** Solar cell or PV cell is made of thin wafers of semiconductor material like silicon that forms the semi-conducting N-P junction, corresponding to negative and positive electrodes. The N-and P-type materials are obtained by doping silicon with N-type and P-type impurities.

There are two types of **dopants** used in doping the tetravalent silicon:

- Pentavalents like arsenic (As), antimony (Sb) or phosphorus (P)
- Trivalents like Indium (In), Boron (B) or Aluminium (Al).

When we dope Si with a pentavalent element, four of its electrons bond with four silicon neighbours, whereas the fifth excess electron is free to move due to loose bonding. Here, the number of conduction electrons are more than the number of holes. Hence the majority charge carriers are negatively charged electrons and therefore known as **N-type semiconductors**.

On the other hand, when tetravalent silicon is doped with a trivalent element, the dopant has one outer electron less than Si. Therefore, this atom can form bonds on three sides with Si, but fails to form bond on one side. In order to hold the dopant atom (*e.g.* boron), tightly within the crystal lattice of Si, some of the electrons bond on outer side in the neighbouring area tend to slide into this vacant bond, leaving a 'hole' at its own site. This hole is available for conduction. Here the holes are the majority carriers while electrons are minority carriers.

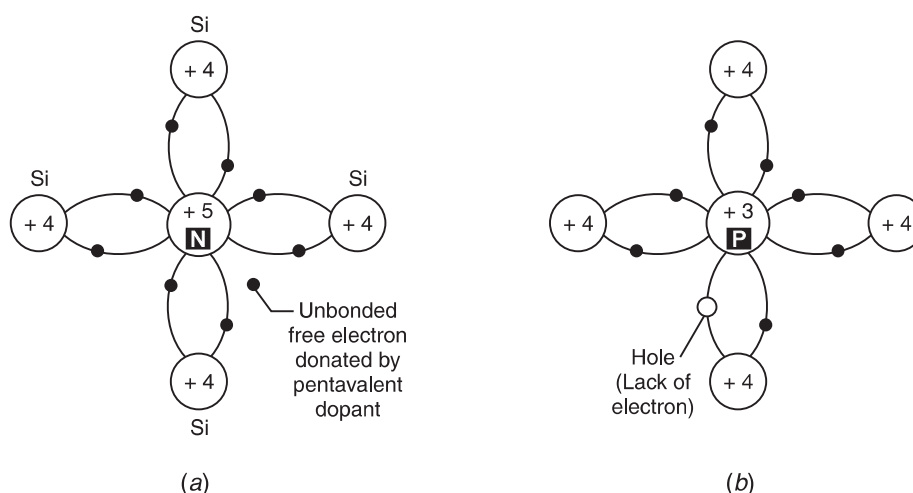


Fig. 5.5. (a) Pentavalent donor atom doped for N-type semiconductor
(b) trivalent acceptor atom doped for P-type semiconductor.

As the solar radiations strike the thin transparent N-type layer, some of the radiations penetrate upto the thick P-type layer. Photons present in the light radiations result in liberation of electron-hole pairs in the P-N junction. Electrons (negative charge) are released from N-type semiconductor and holes (positive charge due to lack of electrons) are created in the **P-type semiconductor** (Fig. 5.5). The potential difference causes flow of electrons, when the electric circuit is completed by connecting electrodes to the load. Thus, there is direct conversion of solar energy to electrical energy.

Silicon used in PV cells can be obtained from silica or sand, which is abundantly available and inexpensive. The potential difference produced by a single PV cell of 4 cm² size is about 0.4–0.5 volts, produces a current of 60–75 milli amperes, and has a rated power of about 0.3 watts. Figure 5.6 shows the structure of a solar cell.

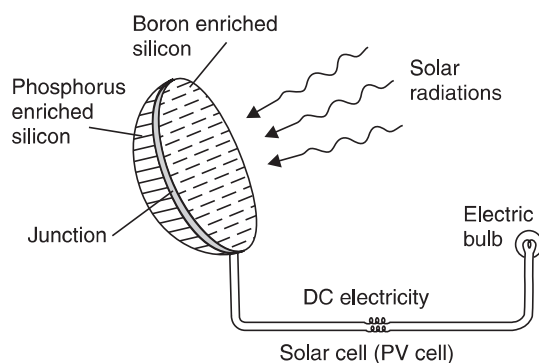


Fig. 5.6. Solar cell

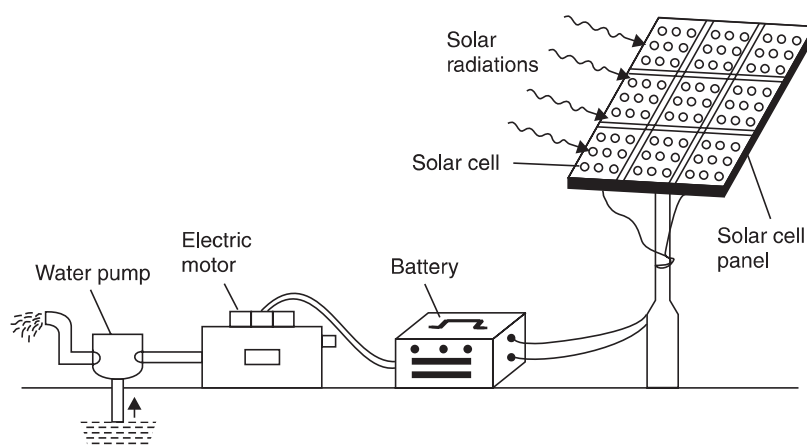


Fig. 5.7. A solar pump run by electricity produced by solar cells.

A group of solar cells joined together in a definite pattern form a solar panel which can harness a large amount of solar energy and can produce electricity enough to run street-light, irrigation water pump etc. (Fig. 5.7).

Solar cells are widely used in calculators, electronic watches, radios, street lighting, traffic signals, water pumps etc. They are also used in artificial satellites for electricity generation.

(iii) **Solar cooker:** Solar cookers make use of solar heat by reflecting the solar radiations using a mirror directly on to a glass sheet which covers the black insulated box within which the raw food is kept as shown in Fig. 5.8. A new design of solar cooker is now available which involves a spherical reflector (concave or parabolic reflector) instead of plane mirror that has more heating effect and hence greater efficiency.

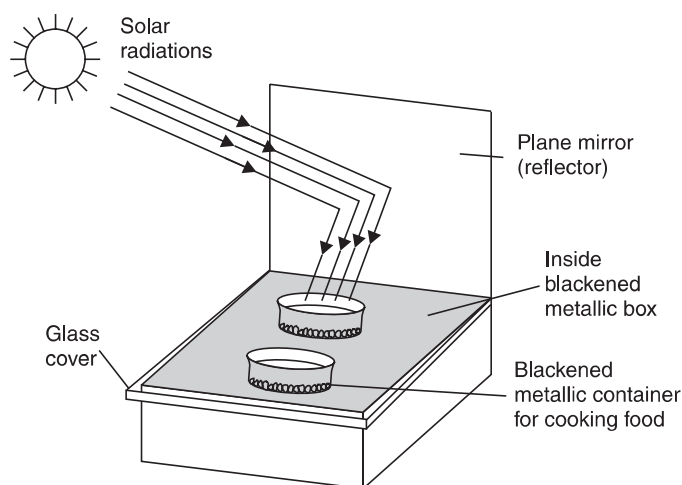


Fig. 5.8. Simple box-type solar cooker.

The food cooked in solar cookers is more nutritious due to slow heating. However, it has the limitation that it cannot be used at night or on cloudy days. Moreover, the direction of the cooker has to be adjusted according to the direction of the sun rays.

(iv) **Solar water heater:** It consists of an insulated box painted black from inside and having a glass lid to receive and store solar heat. Inside the box it has black painted copper coil through which cold water is made to flow in, which gets heated and flows out into a storage tank. The hot water from the storage tank fitted on roof top is then supplied through pipes into buildings like hotels and hospitals.

(v) **Solar furnace:** Here thousands of small plane mirrors are arranged in concave reflectors, all of which collect the solar heat and produce as high a temperature as 3000°C .

(vi) **Solar power plant:** Solar energy is harnessed on a large scale by using concave reflectors which cause boiling of water to produce steam. The steam turbine drives a generator to produce electricity.

5.5.3 Biomass Conversion Processes

Energy locked in biomass can be converted into useful energy by following methods:

(i) **Incineration or direct combustion:** The organic matter like dry municipal waste, bagasse, farm waste, wood chips etc., are burnt at high temperature in incinerators to produce heat, steam and electricity simultaneously, and hence it is called cogeneration.

(ii) **Thermochemical conversion:** The organic matter is decomposed through thermochemical processes having different temperature and pressure combinations.

Biomass can be converted into gases, through a process called **gasification**, by heating with limited air or heating at high temperature and pressure in the presence of steam and oxygen. Biomass can also be converted into gases, liquids or solids through **pyrolysis** at high temperature ($500 - 900^{\circ}\text{C}$) in the absence of oxygen. A large variety of products can be obtained from biomass through pyrolytic destructive distillation.

(iii) **Biochemical conversion:** It may involve anaerobic digestion or fermentation of biomass using microbes to yield biogas and alcohols.

5.5.4 Biogas Production

Biogas plants used in our country are basically of two types:

1. Floating gas-holder type and 2. Fixed-dome type.

1. Floating gas holder type biogas plant: This type has a well-shaped digester tank which is placed under the ground and made up of bricks. In the digester tank, over the dung slurry an inverted steel drum floats to hold the bio-gas produced. The gas holder can move which is controlled by a pipe and the gas outlet is regulated by a valve. The digester tank has a partition wall and one side of it receives the dung-water mixture through inlet pipe while the other side discharges the spent slurry through outlet pipe (Fig. 5.9).

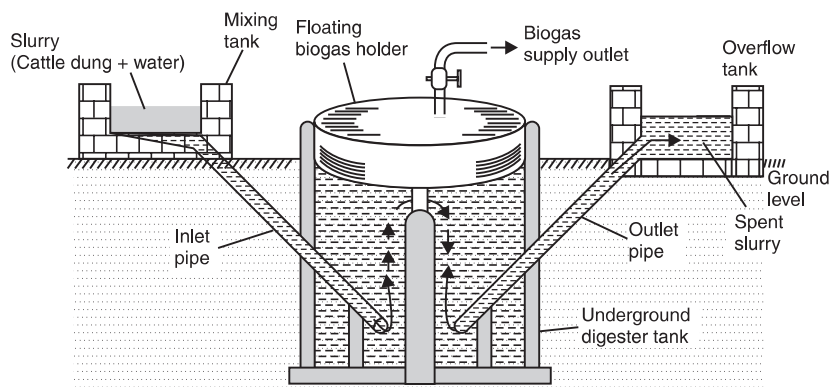


Fig. 5.9. Floating gas holder type biogas plant.

Sometimes corrosion of steel gas-holder leads to leakage of biogas. The tank has to be painted time and again for maintenance which increases the cost. Hence, another type was designed as discussed below:

2. Fixed dome type biogas plant: The structure is almost similar to that of the previous type. However, instead of a steel gas-holder there is dome shaped roof made of cement and bricks. Instead of partitioning, here there is a single unit in the main digester but it has inlet and outlet chambers as shown in Fig. 5.10.

The Ministry of Non-Conventional Energy Sources (MNES) has been promoting the biogas programme in India. Out of the various models, the important ones used in rural set-up are KVIC Model (Floating drum type), Janta Model (Fixed dome type), Deenbandhu Model (Fixed dome type), Pragati Model (Floating drum type), Ganesh Model (KVIC type but made of bamboo and polythene sheet) and Ferro-cement digester Model (KVIC type with ferro-cement digester).

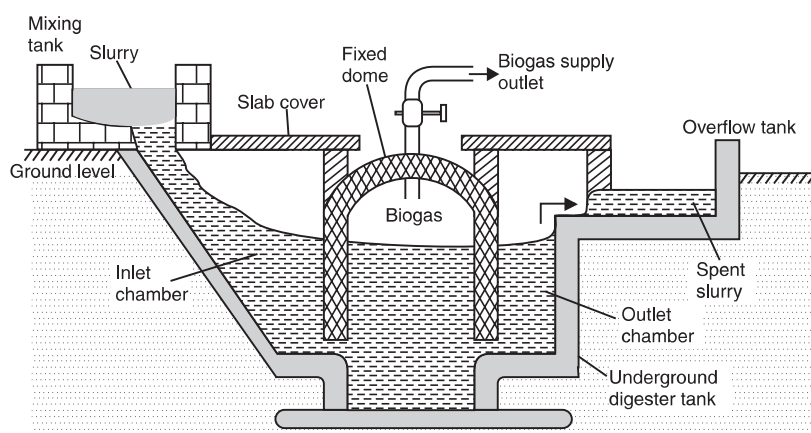


Fig. 5.10. Fixed dome type biogas plant.

5.5.5 Biofuels

Biofuels can be obtained by fermenting biomass that produces alcohols like ethanol and methanol. **Ethanol** can be easily produced from carbohydrate rich substances like sugarcane, corn and sorghum (Jowar). It burns clean and is non-polluting. However, as compared to petrol its calorific value is less and therefore, produces much less heat than petrol. It is also considered to be an excellent substitute for kerosene and its combustion is as clean as LPG. Ethanol is obtained from grain-based or sugar-containing plants like maize, cereals or even organic wastes.

Methanol is very useful since it burns at a lower temperature than gasoline or diesel. Thus the bulky radiator may be substituted by sleek designs in our cars. Methanol too is a clean, non-polluting fuel. Methanol can be easily obtained from woody plants.

Gasohol is a common fuel used in Brazil and Zimbabwe for running cars and buses. In India too gasohol is planned to be used on trial basis in some parts of the country, to start with in Kanpur. Gasohol is a mixture of ethanol and gasoline.

5.5.6 Hydrogen as a Fuel

As hydrogen burns in air, it combines with oxygen to form water and a large amount of energy (150 kilojoules per gram) is released. Due to its high, rather the highest calorific value, hydrogen can serve as an excellent fuel. Moreover, it is non-polluting and can be easily produced.

Production of hydrogen is possible by thermal dissociation, photolysis or electrolysis of water:

- (i) By thermal dissociation of water (at 3000 K or above) hydrogen (H_2) is produced.
- (ii) Thermochemically, hydrogen is produced by chemical reaction of water with some other chemicals in 2–3 cycles so that we do not need the high temperatures as in direct thermal method and ultimately H_2 is produced.
- (iii) Electrolytic method dissociates water into hydrogen (H_2) and oxygen by making a current flow through it.

(iv) Photolysis of water involves breakdown of water in the presence of sunlight to release hydrogen. Green plants and micro-algae also carry out photolysis of water during photosynthesis. Efforts are underway to trap the hydrogen molecule which is produced during photosynthesis. Hydrogen generated by microbial systems is called **biohydrogens**.

However, hydrogen is highly inflammable and explosive in nature. Hence, safe handling is required for using H_2 as a fuel. Also, it is difficult to store and transport. And being very light, it would have to be stored in bulk.

Presently, H_2 is used in the form of liquid hydrogen as a fuel in spaceships. H_2 can be used in **fuel cell** to generate electricity. In fuel cell hydrogen is burnt in air or oxygen in the presence of an electrolyte to produce electricity.

I. QUESTIONS

1. What are renewable and non-renewable resources? Give examples.
2. Give an account of non-renewable energy resources?
3. What are electromagnetic radiations? What are the anthropogenic sources of these radiations?
4. Discuss properties of electromagnetic radiations.
5. What are solar cells? Draw a diagram of a solar cell and enumerate its applications.
6. What is biogas? Discuss the structure and function of biogas plants.
7. What is nuclear energy? Discuss its two types.
8. What is hydro-electric energy? How is it generated?
9. Can hydrogen be used as a fuel? How can hydrogen be produced?
10. Discuss various benefits of using hydrogen as a fuel.

II. OBJECTIVE TYPE QUESTIONS

(A) FILL IN THE BLANKS

1. resources are inexhaustible resources which can be generated within a given span of time.
2. resources can not be generated.
3., and are the natural sources of electromagnetic radiations.
4., and are man made sources of electromagnetic radiations.
5. is the most abundant fossil fuel in the world.
6., and are conventional sources of energy.
7. Natural gas has high calorific value of about kJ/g.
8. Biogas is produced by degradation (in the absence of oxygen) of biological wastes.
9. 95% natural gas is
10. Gasohol is a mixture of and

11. Nuclear energy by nuclear fission is generated when certain isotopes are bombarded by
.....
12. Electrolytic method dissociates water into and

(B) CHOOSE THE CORRECT ANSWER

- Which of the following is included under non-conventional source of energy?
(a) Hydroelectric energy (b) Solar energy
(c) Hydrogen (d) All of these.
- Which of the following are the sources of electromagnetic radiations?
(a) Thunder storms (b) Electric gadgets
(c) Mobile phones (d) All of these.
- Biomass energy can be obtained from
(a) Energy plantations (b) Petro crops
(c) Agricultural and urban waste biomass
(d) All of these.
- Which of the following types of coal has maximum carbon and calorific value?
(a) Anthracite (hard coal) (b) Bituminous (soft coal)
(c) Lignite (brown coal) (d) Wood coal.
- Nuclear energy can be generated by
(a) Nuclear fusion (b) Nuclear fission
(c) Both of these (d) None of these.

(C) WRITE TRUE OR FALSE

- It takes millions of years for formation of coal from trees. (True/False)
- Electromagnetic radiations can travel in space without any material medium. (True/False)
- Electromagnetic radiations have a wide range of wavelength. (True/False)
- Solar cells are made up of thin wafers of semi-conductor materials like silicon and gallium. (True/False)
- Solar heat can not be used to operate street lights, water pumps, television, calculators etc. (True/False)
- Ideal location for installation of wind-mills are coastal regions, open grasslands, hilly regions. (True/False)
- Hydropower also causes environmental pollution. (True/False)
- Burning of dung produces biomass energy but doesn't destroy essential nutrients like N and P. (True/False)
- Sludge left over in the biogas plant cannot be used as a fertilizer. (True/False)
- Ethyl mercaptan (a foul smelling gas) is added to odourless LPG for instantaneous detection of any leakage. (True/False)